

Week 1 — Full Lesson

"Build Your First App in One Sitting" — A Patient-Education Quiz

*A complete, step-by-step lesson for total beginners. No coding. No patient data.
Companion to the Vibe Coding for Healthcare Professionals syllabus.*

A note on the screenshots: The boxes labeled **[SCREEN]** below are *described mockups* of what you'll see — drawn in text so they stay accurate even as the tools update their visuals. Real apps change their buttons and colors often; the *labels and actions* described here are what to look for. When the wording differs slightly, match the closest button.

Lesson at a Glance

- **Goal:** Build and publish a live "Diabetes Basics" patient-education quiz.
- **Time:** 60–90 minutes.
- **Tool:** Lovable (no installation, runs in your web browser).
- **Cost:** A free trial is enough to finish this lesson.
- **You'll leave with:** A real web link you can open on your phone and share.
- **Data safety:** Zero patient data. Everything here is public health-education content.

What you need

- A laptop or desktop with a web browser (Chrome, Edge, or Safari).
 - An email address.
 - 90 quiet minutes.
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Part 0 — The Safety Check (2 minutes, every time)

Before *any* build, say this out loud:

"Does this tool touch real patient information? "

Our quiz uses **general public health facts** about diabetes. **No patient names, no records, no PHI.** Safe to build.

If the answer is ever "yes, it touches real patient data," you **stop** and talk to IT/compliance. For this lesson, we are firmly in the **green zone**.

Part 1 — Create Your Account (5 minutes)

Step 1. Open your browser and go to **lovable.dev**.

Step 2. Click **Sign up** (top-right). Use your email, or "Continue with Google."

[SCREEN] — Lovable home page

```
+-----+
| Lovable                                [ Log in ] [ Sign up ] |
|                                                                              |
|      Build something Lovable                                              |
|      Describe your idea. Get a working app.                              |
|                                                                              |
|      [ Describe what you want to build... ]                               |
|      [                               ( ↑ Send ) ]                       |
+-----+
```

Step 3. Confirm your email if asked. You'll land on a big text box that says something like *"Describe what you want to build."* That box is where the magic happens.

Teaching script: "See that one text box? That is the entire 'programming language' we need today — plain English. We're going to type a description, and the AI builds the app. If you can write a patient-instruction sheet, you can do this."

Part 2 — Plan Before You Build (10 minutes)

Good builds start with a clear request. We'll write our description in three simple parts.

The 3-part recipe for any app prompt:

1. **What it is** — one sentence.

2. **What's in it** — the pieces/content.
3. **How it looks & behaves** — style and buttons.

Here is our finished plan for the quiz. **Don't type it yet** — just read it:

What it is: A simple patient-education quiz app called "Diabetes Basics."

What's in it: A welcome screen with a "Start Quiz" button, then 5 multiple-choice questions (one at a time) about type 2 diabetes, each with 3 answer options. After each answer, show whether it was right and a one-sentence explanation. At the end, show the score out of 5 and a "Try Again" button.

How it looks & behaves: Clean and friendly, large readable text, calm blue-and-white colors, big buttons suitable for older patients. Mobile-friendly.

Teaching script: "Notice we never said *how* to build it — no code words. We described the *experience* a patient would have. That's the whole skill."

Part 3 — Generate Version 1 (10 minutes)

Step 4. Click into the big text box. Copy the **prompt script** below and paste it in.

Prompt Script #1 — The First Build

Build a simple patient-education quiz web app called "Diabetes Basics."

Content:

- A welcome screen with the title, one friendly sentence, and a big "Start Quiz" button.
- 5 multiple-choice questions about type 2 diabetes, shown one at a time.
- Each question has 3 answer options.
- After the user picks an answer, show if it was correct and a one-sentence explanation.
- A "Next" button moves to the next question.
- At the end, show the score out of 5 and a "Try Again" button that restarts the quiz.

Style:

- Clean, friendly, and calm. Blue and white colors.
- Large, easy-to-read text and big buttons, suitable for older adults.
- Works well on a phone screen.

Use accurate, general, public health-education facts about type 2 diabetes.

Do not include any real patient information.

Step 5. Press **Send** (the arrow / "Send" button). Now wait — the AI writes the app while you watch.

```
[SCREEN] — Building
+-----+-----+
| Chat (your messages) | Live Preview |
|                       |               |
| You: Build a simple... | Generating your app... |
|                       | • creating welcome screen |
| Lovable: Working on it | • adding 5 questions |
| ✓ Created welcome      | • styling buttons |
| ✓ Added questions      |               |
| ✓ Added scoring        | [ preview appears here ] |
+-----+-----+
```

Step 6. In 1–3 minutes, a working app appears in the **Live Preview** on the right. Click **Start Quiz** and try it.

Teaching script: "Congratulations — you just built software. Don't worry if it's not perfect. First versions never are. Our real job starts now: testing and refining."

Part 4 — Test Like a Skeptic (10 minutes)

Your job is to *try to break it*. Click everything. Use this checklist:

Week 1 Test Checklist

- [] Does **Start Quiz** open the first question?
- [] Are all **5 questions** there?
- [] Does each question have **3 options**?
- [] When I pick an answer, does it tell me **right or wrong**?
- [] Is there a **one-sentence explanation** after each answer?
- [] Does **Next** move forward every time?
- [] At the end, is there a **score out of 5**?
- [] Does **Try Again** restart from the beginning?
- [] On a **narrow window** (drag it skinny like a phone), does text still fit?

Write down anything that's wrong or ugly. Each item becomes a refinement in Part 5.

Teaching script: "Testers aren't pessimists — they're the patient's advocate. If a confused click can happen, a real patient will find it. Note it; we'll fix it."

Part 5 — Refine in Small Steps (15 minutes)

Now we improve the app **one change at a time**, typing each request into the same chat box. After each one, the preview updates — re-test before the next change.

Prompt Scripts — Common First Refinements

Use only the ones you actually need:

Make text bigger / friendlier:

Make all the text larger and add more space between the buttons, so it's comfortable for older adults to read and tap.

Add a progress indicator:

At the top of each question, show "Question 2 of 5" so users know how far along they are.

Improve the ending:

On the final score screen, add an encouraging message: if the score is 4 or 5, say "Great job!"; if it's 3 or lower, say "Good start — review and try again." Keep the "Try Again" button.

Fix a specific bug (example):

When I click "Next" on question 3, nothing happens. Please make the "Next" button move to the following question every time.

Teaching script — the #1 habit: "One change, then test. If you ask for five things at once and something breaks, you won't know which one caused it. Slow is smooth, and smooth is fast."

Golden rule of bug reports: Describe **what happened**, not how to fix it. "When I click Save, nothing happens" gives the AI everything it needs. You don't have to know the solution.

Part 6 — Publish It (10 minutes)

Step 7. Find the **Publish** (or **Deploy**) button — usually top-right.

[SCREEN] — Top bar

```
+-----+
| Diabetes Basics           [ Share ] [ Publish ► ] |
+-----+
```

Step 8. Click **Publish**. After a few seconds you'll get a **live link** like `diabetes-basics.lovable.app`.

Step 9. Open that link on your **phone**. It's a real, live app on the internet now.

Step 10. Send the link to one trusted colleague and ask: *"Would this help a newly diagnosed patient? What's confusing?"*

Teaching script: "That link works on any phone, anywhere — you didn't install a thing. This is the moment most people realize they can actually do this."

Part 7 — Wrap-Up & Homework (5 minutes)

You did it. You planned, built, tested, refined, and published a real app — with zero code and zero patient data.

Homework for next session

1. **Re-test** your published quiz on a phone and fix **one** thing you don't like (one small prompt).
2. **Write a plan** (the 3-part recipe) for a tool *you* want — a checklist, a calculator, a flashcard set. Don't build it yet; just write the description.
3. **Start your Prompt Library:** open a notes file and paste in any prompt that worked well today.

Self-check: did you meet the goals?

- ☐ I created an account and found the "describe your app" box.
- ☐ I wrote a clear 3-part plan.
- ☐ I generated a working first version.
- ☐ I tested it against the checklist.
- ☐ I made at least one refinement, one change at a time.
- ☐ I published it and opened the live link on my phone.
- ☐ I confirmed: **no real patient data** anywhere.

Quick Troubleshooting

Problem	What to do
The app didn't generate / errored	Click Send again, or paste: <i>"That didn't work. Please try building the quiz again."</i>
A button does nothing	Tell the AI exactly: <i>"When I click X, nothing happens. Please make X do Y."</i>
It looks cramped on a phone	Paste: <i>"Make it mobile-friendly with larger text and spacing."</i>
It changed something I liked	Paste: <i>"Undo the last change and go back to how it was before."</i>
I'm out of free credits	You can stop here — your published link still works. Upgrading is optional.

Instructor Notes (for whoever is teaching this)

- **Group size:** Works 1-on-1 or up to ~12 with one helper roaming.
 - **Pre-class setup:** Ask learners to create the account *before* class to save 5 minutes.
 - **The one rule to over-emphasize:** the data-safety check in Part 0. Repeat it at the start and end.
 - **Common stumble:** learners try to fix things themselves instead of *describing the problem*. Coach them back to "describe what happened."
 - **If someone finishes early:** have them add a 6th question, or restyle with the prompt *"Give it a warmer, more reassuring look."*
 - **Time-saver:** keep Prompt Script #1 on a handout or shared doc so nobody has to retype it.
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This lesson is educational and uses only public health-education content. It is not legal or compliance advice. Never enter real patient information into AI tools, and follow your organization's policies.